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PROPOSAL · PREPARED FOR ZUCAR IT SOLUTIONS

Water Billing Platform — Technical & Commercial Proposal

A multi-tenant water billing and revenue-collection platform for ZUCAR IT Solutions and the utilities it serves.

```
// built in Kenya, for Kenyan utilities
// M-Pesa, SMS and offline meter reading from day one
// one platform, many utilities
```

DOC	VERSION	DATE
MB-10	1.0 draft	08 Oct 2026

STATUS	AUDIENCE	CLASS
For discussion	ZUCAR IT Solutions — management and technical	Commercial in confidence

Prepared by 404HubSpot · Nakuru, Kenya · https://404-hub-spot.vercel.app
Legal entity name and registration details to be inserted before issue. Valid for 30 days from the date shown.

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// 1. Executive summary

ZUCAR has a clear market opportunity: water utilities, estates and institutions across Kenya need accurate billing, fast collection, and trustworthy reporting. 404HubSpot proposes to design, build and operate a **multi-tenant water billing platform** that ZUCAR can offer to many customers, starting with one utility and scaling county by county.

- **One platform, many utilities.** Each customer (tenant) gets isolated data, its own tariffs, branding and users, on shared, cost-efficient infrastructure.
- **Kenya-first.** M-Pesa Paybill reconciliation, SMS bills and receipts, and an offline meter-reading app are part of the core plan, not afterthoughts.
- **Ready for scale.** Designed so a Nairobi-sized utility can run on a dedicated deployment without a rewrite, while small estates stay affordable.
- **Flexible commercial paths.** ZUCAR can resell the platform, license dedicated deployments for large customers, or discuss acquiring rights under a separate agreement.
- **Trustworthy by design.** Immutable financial ledger, full audit trail, role-based access with approvals, and data protection built to Kenyan law.

At a glance	
Delivery	Pilot go-live targeted about 22 weeks from kick-off (see section 6)
First customer	ZUCAR’s first utility, as agreed in discovery
Scope	Standard package from ZUCAR’s presentation, plus the Kenya-essential extensions in section 4
Operating model	Hosted and operated by 404HubSpot, with ZUCAR as partner; dedicated or client-owned deployment available
Investment	Setup fee + per-connection subscription; details in section 8

// 2. Our understanding

ZUCAR’s presentation describes a Standard Water Billing Software covering customer registration, meter management, meter readings, tariff-based billing, payments and receipts, customer statements, reports, and administration, with M-Pesa, SMS, mobile meter reading and customer self-service as possible enhancements.

Today’s challenge	Effect on the utility	What the platform changes
Manual or spreadsheet billing	Errors, disputes, slow bills	Rule-based billing from approved tariffs, with review and approval before issue
Paper meter reading	Lost sheets, estimated bills, unverifiable reads	Offline mobile capture with photo and GPS evidence, anomaly flags
Manual payment matching	Unposted payments, angry customers, leakage	Automatic M-Pesa posting, suspense queue for exceptions, daily reconciliation
No audit trail	Fraud risk, weak accountability	Tamper-evident audit of every sensitive action
Late, inconsistent reporting	Poor decisions, regulatory friction	Live dashboards and regulator-aligned indicators
Growth needs	Each new utility means a new project	Onboard a new tenant in days, not months

// 3. Proposed solution

The platform is organised so ZUCAR, its customers and 404HubSpot each have a clear place.

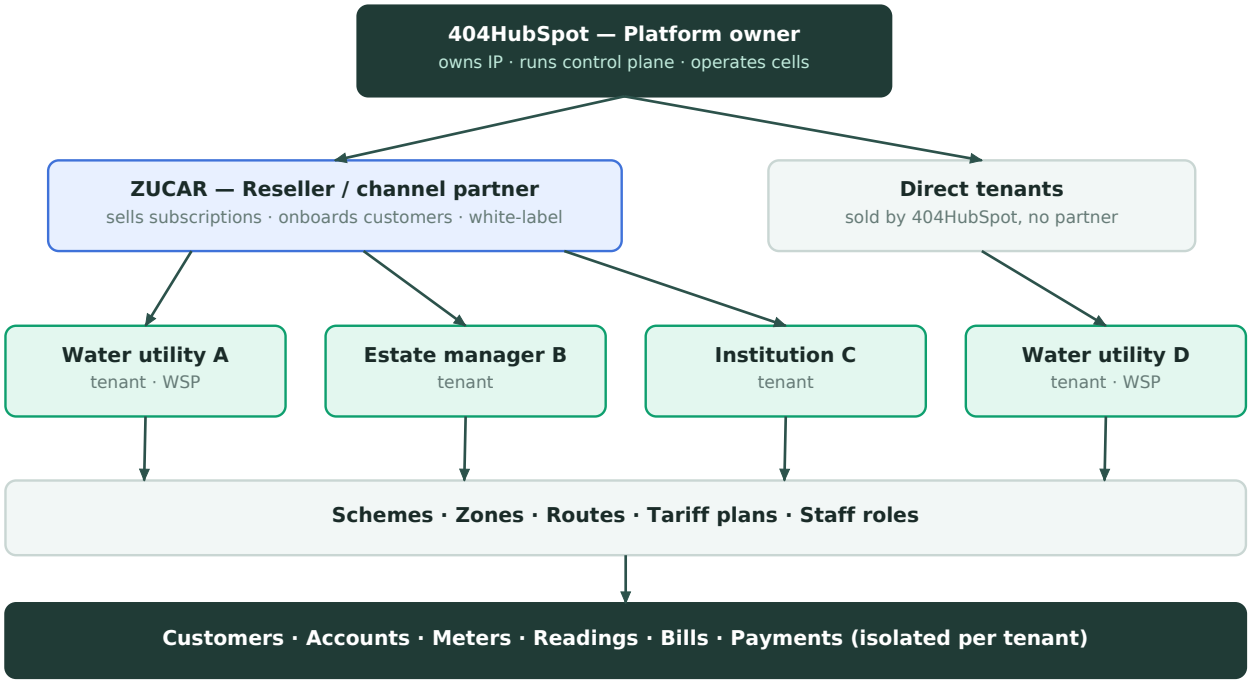


Fig 3.1 404HubSpot builds and operates the platform; ZUCAR resells and supports; each utility is an isolated tenant with its own customers and meters.

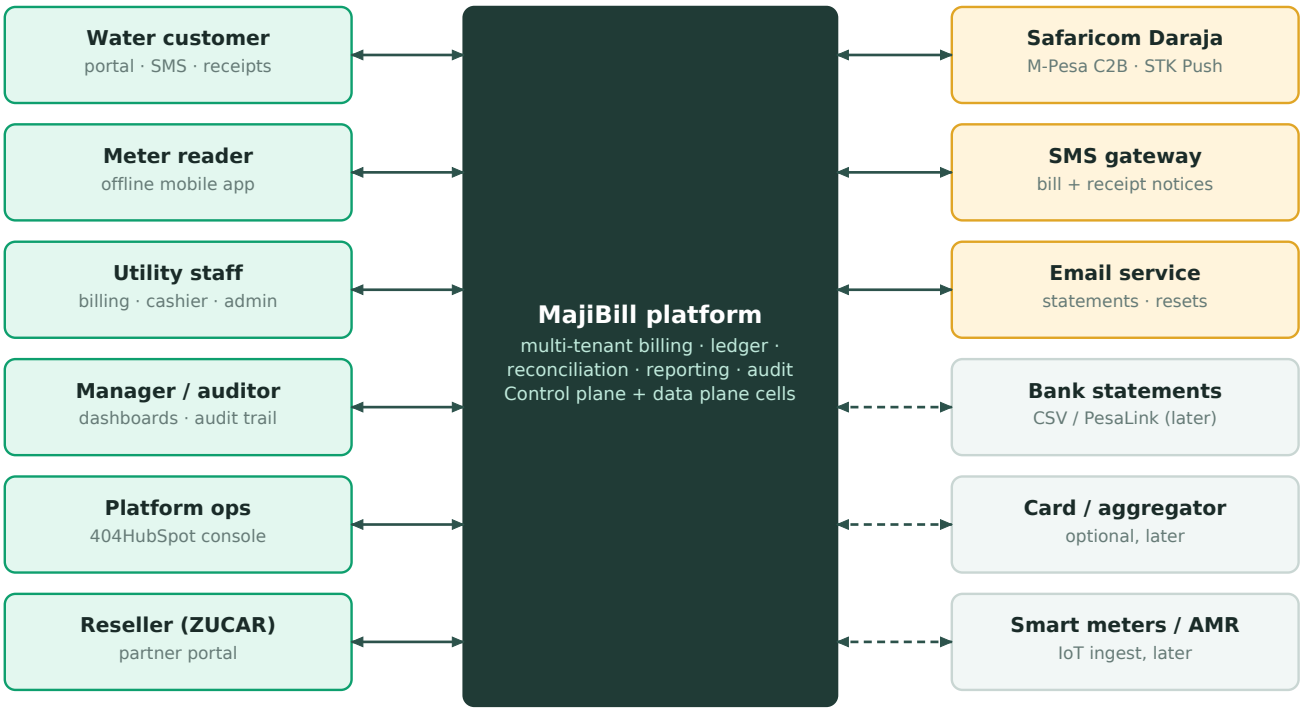


Fig 3.2 Who uses the platform and what it connects to. Dashed items are planned integrations.

3.1 Key capabilities

Capability	What the utility gets
Customers and accounts	Register customers, accounts and meters; status lifecycle; search; bulk import from existing records
Meter reading	Offline mobile app for readers; route assignment; validation and anomaly detection; supervisor review
Tariffs and billing	Block tariffs, fixed and service charges, penalties, rebates; effective-dated tariff changes; bill runs with approval
Payments	M-Pesa Paybill auto-posting, cash and bank payments, receipts, suspense handling, reconciliation
Notifications	SMS bills, receipts and reminders with the utility's own sender name; customer portal
Reports	Arrears, collections, consumption, revenue, exceptions, and regulator-aligned performance indicators
Administration	Users, roles and permissions, maker–checker approvals, branding, templates, full audit viewer

3.2 Why multi-tenant works for ZUCAR

- Every new customer is configuration, not a new build: lower cost per utility and faster time to revenue.
- Improvements ship once to everyone; each utility keeps its own data, tariffs and branding.
- A large utility can be moved to a **dedicated deployment** with the same software if it needs stronger isolation or in-country hosting.
- ZUCAR gets a partner view of its customers' adoption and health, without access to customers' personal data unless the utility agrees.

// 4. Scope and deliverables

Package	Includes	Phase
Standard (from ZUCAR's deck)	Customer registration; meters; readings; tariff-based billing; payments and receipts; statements; reports; user administration	Pilot
Kenya essentials (recommended)	M-Pesa auto-reconciliation; SMS bills and receipts; offline mobile meter-reading app; suspense workflow; maker–checker approvals; audit viewer	Pilot
Platform foundation	Multi-tenant core, platform console for onboarding tenants, partner view for ZUCAR, branding and templates	Pilot
Growth modules	Customer portal with M-Pesa STK push, bank statement import, scheduled reports, SSO, public API and webhooks	After pilot
Advanced	Non-revenue water analytics, anomaly detection, smart-meter integration, additional channels (USSD, email)	Roadmap

4.1 Deliverables

1. Working platform in production for the first tenant, with training and handover.
2. Platform console for tenant onboarding and management.
3. Offline meter-reading mobile app (Android first).
4. Documentation: user guides, administrator guide, API reference, operations runbooks.
5. Data migration of existing customers, meters and balances (scope agreed in discovery).
6. Security testing report and remediation, backup and restore evidence.
7. Pilot report with parallel-run comparison against the utility's existing billing.

4.2 Acceptance

- Bills match the utility's approved calculation for an agreed sample (target: 100% of the sample).
- M-Pesa payments post to the right account within one minute of confirmation in normal operation.
- Reader app completes a full route offline and syncs without data loss.
- A full bill run completes within the agreed window for the pilot population.
- Role and approval rules operate as configured; audit trail shows each sensitive action.

// 5. Approach and team

Work proceeds in two-week sprints with a demo at the end of each. A shared backlog is prioritised with ZUCAR and the utility's finance and operations leads. We use specification-first, agent-assisted engineering with strict human review of financial code, automated tests, and continuous delivery to a staging environment that you can use from the first month.

Role	Responsibility
Technical lead / architect	Architecture, quality, client technical liaison
Backend engineers	Billing, ledger, payments, tenancy, audit
Platform / DevOps engineer	Infrastructure, security, monitoring, releases
Frontend and mobile engineers	Web apps, customer portal, reader app
QA	Test automation, UAT support
Delivery lead	Scope, schedule, reporting, acceptance

Team names and profiles to be inserted at issue. Portfolio: https://404-hub-spot.vercel.app

// 6. Timeline

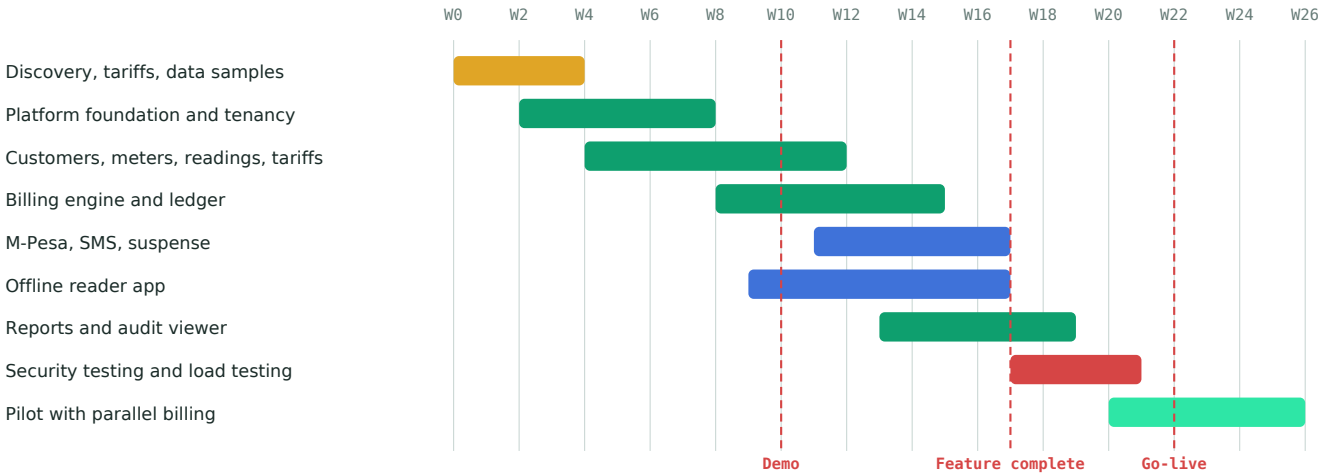


Fig 6.1 Indicative plan. Dates depend on timely access to tariffs, sample data and Safaricom Paybill setup.

Milestone	Target week	Evidence
M0 Kick-off and discovery complete	4	Signed scope, tariffs and data samples received
M1 Core demo	10	Customer to reading to bill to manual payment, on staging
M2 Feature complete	17	M-Pesa sandbox auto-posting, offline reader app, SMS, reports
M3 Pilot go-live	22	Pilot zone live, parallel billing, training done
M4 Pilot acceptance	26	Pilot report approved; roll-out plan agreed

// 7. Security, data protection and reliability

- **Isolation:** each utility's data is separated at the database level; large utilities can have a dedicated deployment.
- **Data protection:** designed for Kenya's Data Protection Act; the utility is the data controller and 404HubSpot the processor under a written agreement; encryption in transit and at rest.
- **Money integrity:** immutable ledger, idempotent payment processing, daily reconciliation, approvals for adjustments and write-offs.
- **Accountability:** tamper-evident audit trail viewable by the utility's auditors; privileged support access is time-limited and visible to the utility.
- **Availability:** target 99.9% monthly availability, tested backups, defined recovery objectives, 24/7 monitoring for large tenants.
- **Testing:** independent penetration test before go-live.

// 8. Commercial proposal

Three ways to work together. All use the same software; they differ in who operates it and who holds which rights. Figures are to be confirmed after discovery.

	A. Reseller partnership	B. Dedicated / licensed deployment	C. Rights acquisition
ZUCAR receives	Right to resell and white-label in an agreed territory	Dedicated or client-hosted deployment for a specific customer	Assignment or exclusive licence for an agreed field of use
Operated by	404HubSpot	404HubSpot or the customer	ZUCAR (with optional support)

	A. Reseller partnership	B. Dedicated / licensed deployment	C. Rights acquisition
Intellectual property	Retained by 404HubSpot; licence to ZUCAR	Retained by 404HubSpot; licence to customer	As agreed in a separate agreement
Commercial shape	Setup fee per tenant + monthly subscription per active connection; partner margin to ZUCAR	Licence fee + setup + annual support	One-off fee, priced separately
Best for	Many small and medium customers	Large utilities, in-country hosting	Full control strategy

8.1 Pricing structure (to be completed)

Item	Basis	Amount
Platform build and first-tenant setup	Fixed, milestone-based (M0, M1, M2, M3, M4)	KES [●]
Per-tenant onboarding	Per tenant: configuration, data migration, training	KES [●]
Platform subscription	Per active billable connection per month, tiered; minimum monthly fee	KES [●] per connection
Partner margin to ZUCAR	Percentage of subscriptions ZUCAR sources	[●] %
Pass-through costs	SMS, payment gateway fees, in-country hosting premium	At cost + agreed handling
Support	Standard / Extended / 24x7 tiers	KES [●] per month
Change requests and new modules	Day rate or fixed quote	KES [●]

// 9. Support and service levels

Tier	Hours	Response (critical)	Includes
Standard	Mon–Fri, business hours (EAT)	4 business hours	Email/ticket support, monthly health report, patch releases
Extended	Mon–Sat, extended hours	2 hours	Plus phone line, quarterly review, priority fixes
24x7	Always on	30 minutes	Plus on-call engineer, incident commander, monthly service review

Critical = system unavailable or payments not posting. Service credits and exclusions to be defined in the service agreement.

// 10. Assumptions and ZUCAR / utility responsibilities

1. Timely access to the utility's approved tariffs, billing rules and sample data in the first four weeks.
2. A named business owner at ZUCAR and at the utility with authority to approve scope and acceptance.
3. The utility obtains and maintains its own M-Pesa Paybill/Till and SMS sender ID, with our guidance; Safaricom approval timelines are outside our control.
4. Utility staff available for training and user acceptance testing.
5. Existing data is supplied in a usable format; clean-up effort beyond the agreed scope is a change request.
6. Hosting region and data-residency requirements are confirmed during discovery.

// 11. Key risks and how we manage them

Risk	Management
Tariff rules or data quality differ from expectations	Early discovery with real samples; flexible, effective-dated tariff model; parallel run before cut-over
Payment integration approval delays	Start applications at kick-off; sandbox-first build; manual posting as fallback
Field adoption of the reader app	Co-design with readers, offline reliability, training and supervisor dashboards
Scope growth	Clear scope baseline, change-control process, roadmap modules sold separately
Data protection compliance	Processing agreement, privacy notices, DPIA, security testing, audit trail

// 12. Terms outline (subject to contract)

- Intellectual property, licence scope, territory and exclusivity as set out in the selected commercial option.
- Each tenant owns its data; platform processes it under a data processing agreement; data export on exit.
- Milestone payments against agreed acceptance criteria; invoicing in KES; taxes as applicable.
- Confidentiality, limitation of liability and service levels as per the master services agreement.
- Change control for scope changes; termination and transition-assistance provisions.
- Governing law: Kenya.

// 13. Next steps

1. Discovery workshop with ZUCAR and the first utility (tariffs, data, workflows, Paybill and SMS).
2. Confirm commercial option and complete section 8 pricing.
3. Agree contract and IP/licence terms; sign.
4. Kick-off and begin Phase 0.

For ZUCAR IT Solutions	For 404HubSpot [registered entity name]
Name:	Name:
Title:	Title:
Signature: Date:	Signature: Date: